

Higher Order Functions

[Start Assignment](#)

- Due Sunday by 11:59pm
- Points 15
- Submitting a file upload

Goal

To have students revise and generalize the higher-order function, *summation*(*n*, *f*), from the course text. Students will also employ an anonymous functions in the form of a lambda expression.

Instructions

Write a higher-order function with the following signature:

```
def summation(f, lower, upper):  
    """This function accepts arguments that include a function, lower bound,  
    and upper bound. It then sums the values from the function for each of  
    the numbers between the lower bound and upper bound, inclusive """  
  
    # revise and update the code from the summation function that was  
    # introduced in section 1.6 of the text
```

Also, be sure to include the following functions:

```
def square(x):  
    """ This function calculates and returns the square of the input argument x """  
  
def fourth_power(x):  
    """ This function calculates and returns the fourth power of x.  
    It uses neither x*x*x*x, x**4, nor math.pow(x, 4) """
```

Finally, have your program use a lambda expression to calculate the sum of the square roots of each of the numbers between the lower bound and the upper bound, inclusive.

The main program should prompt the user to enter the lower and upper bounds, as well as output the results from calling the appropriate functions. The following includes two samples of results from running this program for different sets of input:

```
>>> %Run Assignment3.py  
User Input:  
Enter a lower bound for the sum: 1  
Enter an upper bound for the sum: 10  
  
Program Output:
```

```
The sum of squares of the numbers from 1 to 10 is 385
The sum of the fourth powers of the numbers from 1 to 10 is 25333
The sum of square roots of the numbers from 1 to 10 is 22.4682781862041

>>> %Run Assignment3.py
User Input:
Enter a lower bound for the sum: 2
Enter an upper bound for the sum: 20

Program Output:
The sum of squares of the numbers from 2 to 20 is 2869
The sum of the fourth powers of the numbers from 2 to 20 is 722665
The sum of square roots of the numbers from 2 to 20 is 60.66597781141979
>>>
```

Save the source code week3Lab.py Take the all screenshot and save in to Microsoft word file name Wk3LabScreenshot.doc

Upload the above source code and test program source files and Wk3LabScreenshot.doc)

Grading

- This assignment is worth 15 points and will be graded using the attached grading rubric.

Helpful Resources

See the `summation(n, f)` higher-order function described in section 1.6 of the text.

- [Submitting your Canvas Assignment](https://community.canvaslms.com/t5/Student-Guide/How-do-I-submit-an-online-assignment/ta-p/503)  [_ \(https://community.canvaslms.com/t5/Student-Guide/How-do-I-submit-an-online-assignment/ta-p/503\)](https://community.canvaslms.com/t5/Student-Guide/How-do-I-submit-an-online-assignment/ta-p/503)

Higher-order Functions

Criteria	Ratings			Pts
Introductory documentation that includes the name and description of the program, and the name of the author.	2 pts Full Marks	1 pts Partial credit	0 pts No Marks	2 pts
Correct definition of the fourth_power function.	2 pts Full Marks - Meets specifications	1 pts Partial credit - it works	0 pts No Marks	2 pts
Correct use of lambda expression	3 pts Full Marks	2 pts Partial credit	0 pts No Marks - it doesn't work or is missing	3 pts
Program output similar to that of sample runs illustrated in assignment.	3 pts Full Marks	2 pts Partial credit	0 pts No Marks	3 pts
Correct definition and use of the revised summation function	5 pts Full Marks	3 pts Partial credit	0 pts No Marks	5 pts
Total Points: 15				